

Copy Number Variation Analysis

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Introduction

Copy number variations (CNVs) are large-scale gains or losses of genomic regions, critical in oncology (tumour aneuploidy), developmental disorders (trisomies, deletions), and population genetics. WGS, exome, or array-based readouts feed segmentation algorithms that partition the genome into constant-copy-number segments.

Prerequisites

Coverage or intensity data (read depth, log R ratio); genome windowing.

Theory

Read-depth CNV calling: normalised read counts in windows scale linearly with copy number. The log-ratio series is smoothed and segmented via **Circular Binary Segmentation (CBS)** (DNAcopy), HMMs (HMMcopy), or Bayesian segmentation.

For tumour sequencing, methods like ASCAT and FACETS jointly estimate purity, ploidy, and allele-specific copy number from B-allele frequencies.

Assumptions

Coverage uniformity (controlled by GC / mappability correction); matched normal for tumour calling (ideal); segments have consistent copy number within.

R Implementation

```
library(DNAcopy)

set.seed(2026)
n_probes <- 1000
chrom <- rep(1:5, each = n_probes / 5)
maploc <- rep(seq(1, 1e7, length.out = n_probes / 5), 5)

# Simulated log ratios with two CNV segments
logR <- rnorm(n_probes, 0, 0.2)
logR[201:300] <- logR[201:300] + 0.8 # duplication on chr2
logR[601:680] <- logR[601:680] - 1.0 # deletion on chr4

cna <- CNA(genomdat = logR, chrom = chrom, maploc = maploc,
           data.type = "logratio", sampleid = "tumor1")
smoothed <- smooth.CNA(cna)
seg <- segment(smoothed, verbose = 0)
seg$output # segment means and boundaries

plot(seg)
```

Output & Results

A segment table with chromosome, start, end, number of probes, and mean log ratio; the simulated duplication and deletion are recovered as distinct segments.

Interpretation

“CNV segmentation of the tumour identified gain on 17q (log2 ratio 0.85) and loss on 13q (log2 ratio -1.1); the 13q deletion spans RB1, consistent with the retinoblastoma diagnosis.”

Practical Tips

- GC bias correction (`QDNaseq`) is essential for low-coverage sequencing; uncorrected data produces spurious segments.
- For tumour studies, always use a matched normal to control germline CNVs and sequencing artefacts.
- Integer copy-number calls require ploidy and purity estimates (`ASCAT`, `FACETS`); log ratios alone give only relative copy number.
- Report CNV frequencies across cohorts with `GISTIC2`, which controls for background aneuploidy.
- For rare-CNV studies, focus on large events (> 100 kb); small events are dominated by noise in array data.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- RNA-seq with `DESeq2` / `edgeR`
- Enrichment analysis (`GSEA` / `ORA`)
- scRNA-seq with `Seurat`